

Exam Question Preparation

Information Retrieval and Text Mining

Edition 2025-2026



Maastricht University

Information Retrieval → Text Mining → Conversational Search

1. Building efficient Search Engines including meaningful relevance ranking & relevance feedback.
2. Understand the meaning of text and build structured representations.
3. Use 1 and 2 to drive reliable and factual conversational multi-hop (re)search including real-time quality monitoring.
4. Combine all in agentic architectures.
5. Teach your own proprietary data (text) to a model.



Precision and Recall

Question: An IR system returns 25 relevant documents, and 15 non-relevant documents. There are a total of 50 relevant documents in the collection.

- a. What is the precision of the system on this search.
- b. What is its recall.
- c. What is the f_1 value.
- d. How can we create a f value that prefers recall over precision.
- e. The Kappa value of the data set against which these queries are tested is lower than 60%, what does this say about the quality of the data set.
- f. If the kappa value is too low, how can we fix that?

True or false and explain why

- a. In a Boolean retrieval system, stemming lowers precision.
- b. In a Boolean retrieval system, stemming lowers recall.
- c. In a Boolean retrieval system, the OR operator lowers recall.

Etcetera: you can ask Generative AI to generate more such questions, also for other Boolean operators and other Information Retrieval measurements.

Search Engines

- Index construction
- Dealing with incomplete or erroneous tokens in the index
- Optimal query processing in Boolean search
- Relevance Ranking models
- Relevance Feedback principles and models

➔ Understand differences and be able to explain how they work and why they solve certain problem better.

True or False? Explain Why

- a. Feature Functions from a Conditional Random Fields model can be based on any kind of feature or property of content, form, shape or other information related to text.
- b. Explain how long-distance relations are dealt with in: 1) a Hidden Markov Model, 2) a Conditional Random Field Models, and 3) a BERT models. For each indicate what the maximum distance of such a long-distance relation is in terms of a model parameter or property.
- c. For basic information extraction, shallow parsing is often good enough. Deep parsing is only needed for more complex operations such as co-reference resolution.

Statistical NLP

- Explain why Naive Bayes Classifiers do not work well for natural language processing.
- How do logarithmic operations prevent floating points underflows in statistical NLP and why do they make computations more efficient?

Co-reference Resolution

- Explain why pronoun and co-reference resolution is important for creating a knowledge graph from the content of a book.
- Explain how Dependency Graphs can be used to resolve pronouns. Give an example.

BERT

- Name and explain the 3 components of a BERT Word Embedding.
- Explain what type of linguistic operations could no longer be performed, for each of the above 3 components, if this component would be removed from the embedding (the other 2 components still present).
- How does BERT deal with synonyms?
- Explain why specialized BERT models such as SpanBert or SentenceBert are better in creating embeddings for longer text than just averaging the embeddings of the individual tokens in a context window.
- What would the final embedding look like, if one would take the average of all embeddings in a million-token document?

Conversational Search

- What is hard about conversations? How to deal with that?
- How to insert knowledge with RAG:
 - different models of RAG: text, KG, vector or attention-based.
 - Where: early, encoder, decoder or late fusion?
 - Need training data or not?
 - How to measure quality.
 - ...

Zipf

Q1 — Zipf, stop words, and the cost of "obvious" preprocessing

A colleague claims: "Removing the 100 most frequent words shrinks the index dramatically, so it is always a good first step before indexing or training an embedding model." From an algorithmic and information-theoretic perspective, explain:

(a) **why** her efficiency claim is essentially correct, and

(b) **why** her "always" is wrong.

Illustrate (b) with two concrete retrieval/NLP situations in which removing high-frequency tokens silently breaks the system, and explain the underlying mechanism.

WordPiece Tokenizer

Q2 — Why subword tokenization (WordPiece/BPE) is the right choice for transformer-based retrievers

Explain **why** subword tokenizers such as WordPiece or BPE became the default for BERT-style retrievers and LLMs, where word-level tokenizers (e.g. NLTK + a fixed vocabulary) dominated classical IR.

Address:

- (i) the out-of-vocabulary problem,
- (ii) the relationship between vocabulary size and softmax/embedding-table cost, and
- (iii) the morphological generalization argument.
- (iv) Then state one situation in which a *word-level* tokenizer is still preferable.

TF-IDF

Q1 — Why does TF-IDF still beat dense embeddings on many narrow corpora?

You train two sentiment classifiers on a 2 000-review corpus: a logistic regression on TF-IDF unigrams+bigrams, and a logistic regression on the mean of frozen sentence-transformer embeddings. The TF-IDF model wins by 3–5 F1 points. Explain — at the level of bias/variance, capacity, and information content — **why** this happens, and under what corpus properties the ranking would flip.

BM-25

Q1 — The role of k_1 and b in BM25, and what happens at the extremes

BM25 ranks documents by

$$\text{score}(q, d) = \sum_{t \in q} \text{IDF}(t) \frac{f_{t,d} (k_1 + 1)}{f_{t,d} + k_1 \left(1 - b + b \frac{|d|}{\text{avgdl}} \right)}$$

Explain in your own words **why** these two parameters exist (i.e. what weakness of plain TF-IDF each repairs), and describe the qualitative behaviour of the ranking at the four corner cases $k_1 \rightarrow 0$, $k_1 \rightarrow \infty$, $b=0$, $b=1$. Which corner case effectively reduces BM25 to binary "contains-the-term" retrieval, and why?

Lexical Gap

Q2 — The lexical gap: why dense retrievers help, and why pure dense retrieval still struggles with rare entities

State precisely what the *lexical gap* is, give two concrete query/document pairs that illustrate it, and explain mechanistically why a dual-encoder dense retriever (e.g. DPR, sentence-transformers) closes part of it.

Then explain why pure dense retrieval can be **worse** than BM25 for queries involving rare named entities, product codes, or very recent terms — and what hybrid (BM25 + dense) architectures actually do about it.

Co-References

Q1 — Why a "first-NP-in-sentence" coreference heuristic fails, and what modern systems do instead

A naive information extractor links every pronoun to the most recent noun phrase. Construct two short discourses for which this heuristic extracts a *factually wrong* relation, then explain — at the level of algorithms — why span-ranking neural coreference resolvers (e.g. SpanBERT

end-to-end coref) handle these cases. What is the central reason coreference is *harder* than POS tagging or NER, despite all three being sequence labelling problems?

Negations

Q2 — Why naive negation handling breaks information extraction, and how dependency parsing fixes it

Consider the sentence: *"The MRI showed no evidence of a fracture, but suggested mild inflammation of the rotator cuff."*

A simple negation-aware extractor flips the polarity of any fact whose trigger occurs after the token *"no"*. Show why this rule produces a wrong knowledge base, and explain how a dependency-parse-based scope detector (e.g. NegEx + a syntactic scope, or Stanza's negation feature) would handle it correctly.

Generalize: what abstract property of natural language does the dependency-parse approach exploit that token-window heuristics cannot?

TopicBERT

Q2 — Why does BERTopic outperform LDA on short text — and what failure modes does it inherit from its components?

BERTopic = sentence embeddings → UMAP dimensionality reduction → HDBSCAN clustering → c-TF-IDF topic descriptors. Explain why this pipeline often dominates LDA on short text (tweets, queries, support tickets), and identify, for each of the four components, one failure mode that BERTopic *cannot* fix because the failure originates upstream.

Conversational Search

Q1 — Why does hybrid (BM25 + dense) retrieval beat either component alone, and why is Reciprocal Rank Fusion the default combiner?

State **why** (BM25, dense) is empirically a near-universal Pareto improvement over either component, and explain at the algorithmic level why naive combiners (linear sum of scores, learned weighted sum) typically *under-perform* RRF in practice. Under what corpus or query distribution can RRF still be beaten?

Metrics for Conversational Search

Q2 — Why are classical IR metrics ($P@k$, MRR, nDCG) insufficient for conversational systems?

A conversational search system answers a user query in natural language, optionally citing retrieved passages. Explain why the retrieval-only metrics from earlier modules — $Precision@k$, $Recall@k$, MRR, nDCG — are necessary but not sufficient for evaluating such a system.

List three additional dimensions you must measure, name a concrete metric for each, and explain *why* the dimension cannot be collapsed into a retrieval metric.

Agents

Q1 — When should you use a rigid RAG pipeline, and when an agentic loop?

You are asked to choose between (A) a rigid RAG pipeline (one retrieval call, one generation call, fixed prompt) and (B) an agentic loop (the LLM plans, calls tools, observes results, iterates until it decides to answer).

Identify five task properties that should drive this choice, explain *why* each property favours one or the other, and describe two failure modes that are *introduced* by the agentic architecture and not present in rigid RAG.

Multi-Agent Architectures

Q2 — Multi-agent orchestration: when does it help, and when does it actively hurt?

Multi-agent systems (planner / researcher / coder / critic) are fashionable. Explain (a) what theoretical advantage decomposition into specialised agents has over a single monolithic agent, (b) three mechanisms by which multi-agent orchestration can produce *worse* results than a single well-prompted agent, and (c) which design patterns — supervisor/worker, debate, critic-and-revise — are best suited to which task.

Agentic Training

Q1 — When is fine-tuning on conversion data better than few-shot prompting?

You can teach a model to convert raw text into a target output format (instructions, JSON tool-calls, structured summaries) either by (A) few-shot prompting at inference time or (B) generating a synthetic dataset of (input, output) pairs and supervised fine-tuning. Identify five conditions under which (B) is strictly preferable to (A), and one condition under which (A) is preferable. Explain the underlying mechanism in each case.

Expect questions on...

- Quantitative measurements
- Search Engine Optimizations
- Text Representation methods
- Data Normalization
- Word embeddings.
- Dealing with negations, co-references & pronouns.
- Information Extraction
- Relation Extraction
- Sentiment Analysis
- Text classification
- Clustering & Topic Modeling
- Integration of Conversational AI and Search Engines
- Various RAG methods
- Agentic Architectures and Search
- ...



What to understand...

- Converting natural language to mathematical concepts (TF-IDF, Vectors, Doc-Term Matrices, Embeddings, ...).
- Search Engines: index, search, ranking
- Problems in language (alignment, ambiguity, homonyms, synonyms, hierarchy, references, negations, structural ambiguity, meaning, discourse, speech acts ...) leading to problems for text analytics.
- Methods for text analysis, ultimately leading to a KG.
- Conversation search: how are search engines, text mining for KG and RAG working together.
- Agentic Architectures around a bare LLM
- WHY? Explain, provide examples, ...



More Questions?

